



A COMMUNITY OF CHOICE

CITY ENGINEER'S OFFICE
2101 O'Neil Avenue
Cheyenne, Wyoming 82001

MEMO

TO: Honorable Mayor Patrick J. Collins, City of Cheyenne
Members of the City Council, City of Cheyenne

FROM: Wesley D. Bay, P.E., City Drainage Engineer

DATE: August 26, 2024

SUBJECT: Professional Service Agreement between the City of Cheyenne and Ayres Associates, for the design of 19th Street Drainage and Intersection Improvements Project.

Please find the attached Professional Service Agreement between the City and Ayres Associates for engineering services related to the design of the 19th Street Drainage and Intersection Improvements Project. The design will entail a new hydraulic structure in Crow Creek along 19th Street; 19th Street re-alignment; reconfiguration of the intersections of Missile Drive and 19th Street, and Missile Drive and Old Happy Jack Road.

Engineering Staff evaluates fee proposals for engineering services based on time and material estimates for the specific project. Those costs are readily tracked and verified during design operations and the City only pays for the actual time expended. We believe this agreement will provide the required services to complete the design portion of the project.

The fees include services for project management, intersection design, topographical surveying, geotechnical investigations, utility investigation, structural design, sanitary sewer design, and plan creation. Funding for this project will come from the 2019 1% Specific Purpose Tax Fund (5th Penny) and a FEMA grant. The total professional services will be a cost not-to-exceed \$712,446. We respectfully request your approval of this professional services agreement.

City Contract No. _____

PROFESSIONAL SERVICES AGREEMENT
for
19th Street Drainage and Intersection Improvements Project

This Professional Services Agreement (“Agreement”) is made and entered into by and between the City of Cheyenne, located at 2101 O’Neil Avenue, Cheyenne, Wyoming 82001 (“City”), and Ayres Associates, located at 3665 JFK Parkway, Building 2, Suite 1, Fort Collins, CO 80525 (“Consultant”). This Agreement shall become effective upon the date of the last signature affixed hereto.

RECITALS

In consideration of the mutual covenants and obligations herein expressed, it is agreed by and between the parties as follows:

1. SCOPE OF WORK

The Consultant agrees to provide services as described for the above-mentioned project, in accordance with the Scope of Work attached hereto as Exhibit A, incorporated by reference and expressly made a part hereof.

2. TIME OF PERFORMANCE

This Agreement shall commence on the date the last required signature is attached hereto and shall continue in full force and effect until the scope of work is completed in accordance with Exhibit A, unless this Agreement is earlier terminated pursuant to its provisions.

3. COMPENSATION

In consideration of the services to be performed pursuant to this Agreement, the Consultant will bill the City and the City agrees to pay Consultant a not to exceed amount of Seven-Hundred Twelve-Thousand Four-Hundred Forty-Six and Zero Cents (\$712,446.00) as stated in Exhibit B.

4. CONSULTANT RESPONSIBILITY

In providing services under this Agreement, the Consultant shall perform in a manner consistent with that degree of care and skill ordinarily exercised by members of the same profession currently practicing under similar circumstances. Upon notice to the Consultant, the Consultant will, without additional compensation, correct those services not meeting such a standard.

5. COMPLIANCE WITH LAWS

The Parties agree that they will perform their obligations as provided in this Agreement in accordance with all applicable laws and ordinances.

6. DRUG-FREE WORKPLACE

In compliance with the Drug-Free Workplace Act of November 1988, the City of Cheyenne has established an Alcohol and Controlled Substance Policy that pertains to alcohol and drug usage by City Employees. All architects, engineers, and other consultants under contract with City of Cheyenne, and their employees and sub-consultants, are required to comply with the provisions of the City's Alcohol and Controlled Substance Policy for drug and/or alcohol usage on City property or other sites occupied by the Consultant while performing the duties and responsibilities of the Agreement. It is the responsibility of the Consultant to familiarize themselves with the requirements of this policy and to inform all their employees and sub-consultants of the requirements and ensure their compliance. If the Consultant, their employees or sub-consultants are found in violation of this policy, the Agreement may be terminated.

7. INDEMNIFICATION/HOLD HARMLESS

The Consultant agrees to indemnify, hold harmless, and defend the City from and against any and all liabilities, claims, penalties, forfeitures, and suits, and the cost and expenses incident thereto, including reasonable attorney's fees, which may hereafter arise as a result the performance of the Consultant's duties, including death or bodily injury to any person, destruction or damage to any property, contamination of or adverse effects on the environment, or any violation of governmental laws, regulations, or orders to the extent caused by (1) the Consultant's breach of any term or provision of this Agreement; or (2) any negligent or wrongful act, error, or omission by the Consultant, or its employees or sub-consultants in the performance of this Agreement. The Consultant acknowledges that it may incur a financial obligation to the City pursuant to the terms of this paragraph.

8. INSURANCE REQUIREMENTS

The Consultant must provide proof of the following insurance coverages:

Commercial General Liability Insurance

For claims arising out of bodily injury, illness, or death, or from damage to or destruction of property of others, including loss or use thereof, with minimum limits of \$1,000,000 per occurrence, \$2,000,000 aggregate for the entire term of the Agreement.

Workers' Compensation

Workers' compensation coverage shall be in effect for the entire term of the Agreement, as required by Wyoming law, for all employees or agents providing services under this Agreement. Consultant shall provide the City with proof of workers' compensation or employer's liability insurance coverage.

Professional Liability Insurance

The Consultant shall provide proof of professional liability insurance or errors and omissions liability insurance in an amount not less than \$500,000 to protect the City from any and all claims arising from the Consultant's negligence in the performance of duties under this Agreement. The City prefers that this liability insurance coverage be provided pursuant to an "occurrence" policy.

If this coverage is provided pursuant to a "claims made" policy:

- a. Consultant shall, concurrently with the execution of this Agreement, provide the City with a certificate of insurance demonstrating that such coverage is or shall be in effect at the time the Consultant begins the provision of services under this Agreement; and
- b. In the event the Consultant's services extend into a future policy period, the Consultant shall, prior to the policy expiration date, provide the City with a new certificate of insurance demonstrating that such coverage is or shall be in effect during all periods of time that Consultant will provide services under this Agreement; and
- c. Consultant shall maintain said "claims made" coverage for a period of five (5) years following the last date that Consultant has provided services under this Agreement; and
- d. In the event the Consultant or the insurer terminates "claims made" coverage prior to the expiration of the periods provided in subparagraphs a., b., or c. of this paragraph, the Consultant shall provide to the City advance written notification of the termination of said coverage and shall provide the City with an endorsement for an extended reporting period ("tail coverage") which shall be in effect for a period of time not less than five (5) years following the last date that Consultant has provided services under this Agreement.

Additional Insurance Information

The Consultant shall name the City of Cheyenne as **an Additional Insured** by endorsement on its insurance policies and shall provide the City with a copy of the endorsements. This requirement does not apply to workers' compensation and professional liability insurance policies.

Consultant shall provide the City with certificates of insurance acknowledging the above-stated coverages prior to beginning any work under this Agreement.

It is understood and agreed that these policies are primary and not contributory. All policies required under this Agreement shall be in effect for the duration of the Agreement. It shall be an affirmative obligation upon Consultant to immediately notify in writing the city risk manager, city clerk, and city attorney of any fact, circumstance, or occurrence that has resulted in or may result in the cancellation of, or substantive change

to any insurance coverage required by this Agreement, and failure to do so shall be construed to be a breach of this Agreement.

If requested by the City, the Consultant shall provide the City with copies of insurance policies and/or policy endorsements listing the City of Cheyenne as an additional insured. The City's failure to request or review such policies, endorsements, or certificates shall not affect the City's rights or Consultant's obligation hereunder.

Any insurance company providing coverage under this Agreement shall have a minimum A. M. Best rating of A- (excellent).

9. MINORITY AND DISADVANTAGED BUSINESS ENTERPRISES 49 C.F.R. Part 26
All parties to this Agreement assure that no person will be excluded from participation in, denied the benefits of, or otherwise discriminated against, in connection with the award and performance of this Agreement on the grounds of age, race, color, disability, national origin, or sex.
10. MONTHLY REPORT
With every monthly billing, Consultant shall provide the City Representative with a written statement of the status of the work with respect to the Scope of Work, time sheets, and work schedule. Failure to provide the required monthly report will delay processing of any payment request until the report is submitted.
11. INDEPENDENT CONSULTANT
The Consultant shall function as an independent consultant for the purposes of this Agreement. The Consultant shall assume sole responsibility for any debts or liabilities that may be incurred by the Consultant in fulfilling the terms of this Agreement. Nothing in this Agreement shall be interpreted as authorizing the Consultant or its agents or employees to act as an agent or representative of or on behalf of the City or to incur any obligation of any kind on the behalf of the City.
12. TAXES
The Consultant agrees to pay all valid taxes, excises, license fees, permit fees, bills, debts, and obligations incurred by and in connection with its operations under this Agreement.
13. DEFAULT
Each and every term and condition in this Agreement shall be deemed to be a material element of the Agreement. In the event either party should fail or refuse to perform according to the terms of this Agreement, such party may be declared in default.

14. REMEDIES

In the event a party has been declared in default, such defaulting party shall be allowed a period of fifteen (15) days within which to cure the default. In the event the default remains uncorrected, the non-defaulting party declaring default may elect to:

- a. Terminate the Agreement and seek damages, which damages shall not exceed the contract amount; or
- b. Treat the Agreement as continuing and require specific performance.

15. TERM AND TERMINATION

The term of this Agreement shall be from the date of execution and continue until the scope of work is completed in accordance with Exhibit A. The parties may by mutual written agreement renew or extend this Agreement. The City may, without cause, and upon thirty days' written notice to the Consultant, terminate this Agreement in whole or in part at any time, for the City's convenience. Upon receipt of such notice, the Consultant shall:

- a. Discontinue all services affected; and
- b. Deliver to the City within five (5) days all data, drawings, specifications, reports, estimates, summaries, and such other information and materials as may have been accumulated by the Consultant in performing this Agreement, whether completed or in process.
- c. In the event of termination for convenience, the City will pay the Consultant for accepted work done and expenses incurred to the date of termination. Such acceptance shall not be unreasonably withheld.

16. ADDITIONAL REMEDIES

In the event the Consultant fails to strictly perform in accordance with this Agreement, the City may elect to correct the deficiencies and charge the Consultant. In the event of default of any of the conditions by either party which shall require the party not in default to commence legal or equitable action against the defaulting party each party shall bear its own costs and expenses, including without limitation, attorney's fees.

17. GOVERNING LAW, JURISDICTION AND VENUE

The construction, interpretation, and enforcement of this Agreement shall be governed by the laws of the State of Wyoming. The courts of the State of Wyoming shall have jurisdiction over any action arising out of this Agreement and over the parties, and the venue shall be the First Judicial District, Laramie County, Wyoming.

18. GOVERNMENTAL IMMUNITY

The City of Cheyenne, along with its officials and employees, does not waive its governmental immunity by entering into this Agreement, except to the extent necessary for the parties to pursue a contract action to clarify or enforce the written terms of the Agreement. Furthermore, the City specifically retains all immunities and defenses available to it as a sovereign or governmental entity pursuant to Wyo. Stat. § 1-39-101, *et*

seq., and all other relevant state and federal law. Designations of venue, choice of law, enforcement actions, and similar provisions should not be constructed as a waiver of governmental immunity. The parties agree that any ambiguity in this Agreement shall not be strictly construed, either against or for either party, except that any ambiguity as to governmental or sovereign immunity shall be construed in favor of governmental immunity.

19. MONITORING ACTIVITIES

The City shall have the right to monitor all activities related to this Agreement that are performed by the Consultant or its sub-consultants. This shall include, but not be limited to, the right to make site inspections at any time and with reasonable notice; to bring experts and consultants on site to examine or evaluate completed work or work in progress; to examine the books, ledgers, documents, papers, and records pertinent to this Agreement; and to observe personnel in every phase of performance of the related work.

20. OWNERSHIP OF DOCUMENTS AND WORK PRODUCT

All deliverables by the Consultant, all work product of the Consultant, all notes, calculations, memoranda, or any other documentation generated by the Consultant pursuant to the scope of this Agreement, shall be the property of the City of Cheyenne and shall be provided to the City of Cheyenne immediately upon request.

21. NON-DISCRIMINATION

The Consultant shall comply with the Civil Rights Act of 1964, the Wyoming Fair Employment Practices Act (Wyo. Stat. § 27-9-105, *et seq.*), the Americans With Disabilities Act (ADA), as amended, 42 U.S.C. § 12101, *et seq.*, and the Age Discrimination Act of 1975 and any properly promulgated rules and regulations thereto, and shall not discriminate against any individual on the grounds of age, sex, creed, color, race, religion, national origin, ancestry, pregnancy or qualifying disability in connection with the performance under this Agreement.

22. FORCE MAJEURE.

The performance of the Agreement by either party shall be subject to force majeure including, but not limited to, acts of God, fire, flood, natural disaster, war or threat of war, acts or threats of terrorism, civil disorder, unauthorized strikes, governmental regulation or advisory, recognized health threats as determined by the World Health Organization, the Centers for Disease Control, or local government authority or health agencies (including, but not limited to, the health threats of COVID-19, H1N1, or similar infectious diseases), curtailment of transportation facilities, or other similar occurrence beyond the control of the parties, where any of those factors, circumstances, situations, or conditions or similar ones prevent, dissuade, or unreasonably delay the performance required by this Agreement. The Agreement may be cancelled by either party, without liability, damages, fees, or penalty, and any unused deposits or amounts paid shall be refunded, for any one or more of the above reasons, by written notice to the other party.

23. PROFESSIONAL REGISTRATION

The Consultant shall endorse, as required by law, plans and reports prepared under this Agreement, and shall affix thereto his or her seal of professional registration, showing that he or she is licensed to practice in the State of Wyoming, if necessary.

24. SUPPLEMENTAL GENERAL CONDITIONS

In consideration of the mutual covenants and obligations herein expressed, it is further agreed by and between the parties as follows:

Standard of Care. The standard of care for all professional engineering and related services performed or furnished by Engineer under this Agreement will be the care and skill ordinarily used by members of the subject profession practicing under similar circumstances at the same time and in the same locality. The Engineer makes no warranties, express or implied, under this Agreement or otherwise, in connection with any services performed or furnished by Engineer.

Technical Accuracy. Owner shall not be responsible for discovering deficiencies in the technical accuracy of Engineer's services. Engineer shall correct deficiencies in technical accuracy without additional compensation unless such corrective action is directly attributable to deficiencies in Owner-furnished information.

Conflicts within the Agreement. If there is a conflict within this Agreement, the order of priority of the documents which make up this Agreement, from highest to lowest, will be:

- (a) Agreement.
- (c) Any supplementary conditions to the General Conditions.
- (d) General Conditions.
- (e) Schedule A – SCOPE OF SERVICES.
- (f) Other schedules to this Agreement.

Any changes in the Consultant's or Subconsultant's staff or fee structure will be presented in writing to the City for approval prior to initiating any changes or performing any work.

Overhead and Normal Business Expenses. The consultant shall include customary overhead costs and statutory benefits as part including, but not limited to, social security contributions, unemployment, excise and payroll taxes, workers' compensation, health and retirement benefits, bonuses, sick leave, vacation, and holiday pay, as applicable, within the billing rates furnished in Schedule B. Additionally, the cost of general and administrative overhead which includes salaries and wages of employees engaged in business operations not directly chargeable to projects, plus non-Project operating costs, including but not limited to, business taxes, legal, rent, utilities, office supplies,

insurance, and other operating costs shall not be directly billed to project and assumed to be included within the billing rates furnished.

Design Process Documentation. To document the design process, the consultant will be required to provide a Decision and Design Log for the project duration. This log shall be furnished to the City of Cheyenne in conjunction with the project meetings. The purpose of the log is to document and track major decisions during the design and development process. The log shall generally follow the format outlined below in Table 1 Design and Decision Log:

Table 1 Design and Decision Log

No.	Date	Decision	Alternatives Considered	Reasons	Decision Made By	Additional Task Required
Sequenced log number.	Date the decision was made.	A one-sentence summary of the decision made.	A list of the other approaches considered.	A two to three sentence summary of why the decision was made. A link to additional documentation, if required and [Title].	Who made this decision?	If applicable, a list of additional tasks required by the design or follow-up required.

25. ENTIRETY OF AGREEMENT

This Agreement consisting of nine (9) pages, Exhibit A consisting of sixteen (16) pages, and Exhibit B consisting of fourteen (14) pages contain the entire understanding of the parties. There are no other terms or conditions, written or oral, concerning or controlling this matter.

IN WITNESS WHEREOF, the Parties to this Agreement, through their duly authorized representatives, have executed this Agreement on the days and dates set out below and certify that they have read, understood, and knowingly and voluntarily agreed to the terms and conditions of this Agreement. This Agreement may be executed in multiple counterparts, each of which shall be deemed an original, but all of which together shall constitute one and the same instrument.

City of Cheyenne

(SEAL)

By: _____
Patrick Collins, Mayor

Attest:

Kristina F. Jones, City Clerk

Date: _____

Consultant

By: _____

Print Name: _____

Date: _____

Exhibit A – Scope of Work

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PROFESSIONAL SERVICES AGREEMENT

Project Name: 19th Street Drainage and Intersection Improvements Project

A.1 Overview

Consultant services for the 19th Street Drainage and Intersection Improvements Project are the design and development of a complete set of construction documents, including plans, specifications, and estimates. Any innovative solutions that are cost effective and meet project goals will be incorporated during project scoping negotiations and included in the design. Work required for this project will include project management; roadway and intersection design; bridge/structural design; hydraulic analysis; geotechnical investigation; storm water and storm sewer design; right-of-way surveying; and utility location, coordination, and design.

The project requires a professional design team that includes a Civil Engineer registered in the State of Wyoming with a minimum of 10 years of project design experience in one or more of the following disciplines: bridge design; urban roadway design; hydraulic analysis; with the ability to provide in-house or capacity to sub-consult for survey services, geotechnical investigation and those disciplines required for the project.

A.2 Project Description

A.2.a Overview

The project is located along 19th Street between Missile Drive and Ames Avenue and extends west of Missile Drive onto Old Happy Jack Road.

The project will include the design of the intersections of 19th Street and Missile Drive and Old Happy Jack Road/Washington Avenue and Missile Drive. Four different intersection configurations were presented in the Metropolitan Planning Organization's (MPO) Missile Drive Corridor Plan completed in 2010. Alternative three was selected as the preferred option. This document can be found at:

https://www.plancheyenne.org/wp-content/uploads/2012/12/MissileDriveFinalReport_100510_acd.pdf

The use of a roundabout was not evaluated in the 2010 corridor study. The MPO recently evaluated a merged double roundabout (peanut roundabout) at the intersection of 19th Street, Missile Drive, and Old Happy Jack Road. This evaluation was presented in the Missile Drive Corridor Concept Plan and is included with this scope of work. A single roundabout and merged double roundabout will be evaluated as potential alternatives to the selected intersection configuration in the 2010 study. This will require evaluating future land uses in the West Edge District of Cheyenne for traffic volume projections and impacts to the 100-year FEMA regulated Special Flood Hazard Area (SFHA). If the floodway is encroached on, a Conditional Letter of Map Revision and Letter of Map Revision will be required demonstrating a no-rise condition and documenting changes to the flood plain mapping.

The preferred intersection configuration should be evaluated and designed using traffic projections included in the Connect 2045 Cheyenne Area Transportation Master Plan.

The project will also include the design of a structure at Crow Creek. Hydraulic analyses will be performed to determine what type of channel and structure clearance is needed to eliminate overtopping at 19th Street. Downstream impacts at Lincolnway and Ames Avenue Underpass will need to be evaluated in the analysis as well. Precast girders, arch structures, and box culverts should be evaluated in the design in order to

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Project Name: 19th Street Drainage and Intersection Improvements Project

expedite construction. Selection of the structure type should consider project construction costs, lifecycle maintenance costs, and minimization of impacts to the travelling public.

A reconfigured Greenway connection at 19th Street from Martin Luther King Jr. Park will be included in the design. Currently, the MPO is in the process of completing the West Crow Creek Greenway Plan that has a potential connection at this location. Coordination and inclusion of this planning effort will be required for the design of the Greenway connection.

A sanitary sewer main is planned to be installed in Missile Drive for the Extended Use Lease development on Warren Air Force Base property. Design coordination will be required with the Board of Public Utilities for the location of the proposed line.

A.2.b Project History

Resolution 5933, which was passed in 2018, identified this as a priority project. The intersection has operational and safety issues due to a severely skewed alignment and also lacks a pedestrian corridor connection. The current culverts in Crow Creek at 19th Street have inadequate conveyance capacity which leads to overtopping and storm water inundation in parts of the downtown area during significant storm events.

A.2.c Existing Facilities

19th Street, 20th Street, Missile Drive, and Dey Avenue are classified as minor arterial roadways. Old Happy Jack is classified as a minor collector. The remaining streets in the area are classified as local streets. 19th and 20th Streets have two lanes and are configured as a one-way couplet, with a short section of 19th Street being two-way east of the intersection. Missile Drive has four through lanes, a center median south of the intersection, an eastbound center left turn lane, and westbound right slip lane north of the intersection. Old Happy Jack has two through lanes and an auxiliary southbound right turn lane at the intersection.

An asphalt Greenway path connects to the south side of 19th Street and then leads west to the intersection and east to a sidewalk. The Greenway terminates at the intersection.

Crow Creek flows through five 14' X 6' corrugated metal pipe arch structures under 19th Street. The structures were installed in 1965. All pipes exhibit moderate corrosion and distortion throughout. The slope paving at the inlet side has failed at multiple locations and the channel is sedimented which directs regular creek flows through only one of the five pipe arches.

A.2.d Proposed Improvements

Proposed improvements will be dictated by the evaluation of a roundabout or the selected alternative presented in the 2010 MPO planning study for the intersections of 19th Street and Missile Drive and Old Happy Jack Road/Washington Avenue and Missile Drive.

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Project Name: 19th Street Drainage and Intersection Improvements Project

Proposed improvements include realignment of Old Happy Jack Road at Missile Drive and W. 19th Street at Missile Drive. Proposed improvements at Westland Road/W. 24th Street & Missile Drive will be dictated by the evaluation of a single-lane roundabout or signalized intersection.

At a minimum, the following should be expected as tasks for the project:

- Roadway and Greenway geometric design.
 - Vertical and horizontal re-alignment of 19th Street.
 - Horizontal re-alignment of Old Happy Jack Road/Washington Avenue.
- Intersection design.
 - Old Happy Jack Rd/Washington St. & Missile Dr. (assumed to be one-way stop control)
 - 19th St. & Missile Dr. (assumed to be one-way stop control)
- Hydraulic design
 - Provide a channel-way and structure sufficient to allow unrestricted flow for the 100-year storm event.
- Structural design
 - Hydraulic structure
 - Retaining walls
- Geotechnical design
 - Shallow and deep foundations
- Pavement design

A.2.e Right-Of-Way

Right-of-way acquisition is not anticipated for this project.

A.2.f Project Funding and Timing

There is sufficient funding allocated toward the design of the project. The channel and structure design will be partially funded by a FEMA grant.

The design phase of the project is anticipated to be completed within ten (10) months from the date of execution of a Professional Services Agreement with the selected firm.

The construction of the project will be postponed until sufficient funding is available. An additional RFP will be released for construction management at that time.

A.3 General Conditions

The following items are included as part of the basic compensation for this project:

- Normal business expenses - phone, postage, etc.
- Cost of insurance.
- In-house computer time and service.
- Word processing, accounting, and man-hour records.
- Permits and license fees.
- Mileage.

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Project Name: 19th Street Drainage and Intersection Improvements Project

- Travel fees, room and board, per diem.
- Printing costs for all standard review, bidding, and as-constructed plans and other correspondence and contract documents.
- Film and processing.

Additional items which will be addressed:

All subconsultant fees at cost, which may include:

- Surveying.
- Utility investigation.
- Geotechnical Investigation.

Other items:

- Overtime engineering and inspection.
- Special inspections, including vacuum excavation potholing.
- Survey materials, stakes, etc.
- Additional required services.

Any changes in the Consultant's or Subconsultant's staff or fee structure will be presented in writing to the City for approval prior to initiating any changes or performing any work.

All pay invoices will include a summary comparing the proposed schedule and actual work completed, actions items outstanding for both the City and Consultant, a summary of work completed during the pay period, and commentary associated with impacts to the schedule.

A.4 Environmental Clearances

This project will require work within the limits of Crow Creek. Coordination and permitting with the US Army Corps of Engineers will be required. See detailed "Environmental Resource Review" scope on the following pages for additional details.

A.5 Public Involvement

Three public meetings will be held for the project. The Consultant will provide public notice and media releases with assistance from the city; coordinate the venue of the meeting; prepare exhibits, handouts, and comment cards; and correlate public comments. A preliminary public meeting will be held to gain public input on the preferred intersection and roadway configuration. Two additional meetings will be held after the 30% and 60% plan issuances.

A.6 General Design

A.6.a Design and Construction Standards

All design and construction will be done in accordance with the latest edition of the applicable standards including, but not limited to, the following:

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Project Name: 19th Street Drainage and Intersection Improvements Project

- AASHTO Policy on Geometric Design of Highways and Streets.
- AASHTO LRFD Bridge Design Specifications.
- AASHTO Roadside Design Guide
- AASHTO Guide for the Development of Bicycle Facilities
- Manual on Uniform Traffic Control Devices.
- City of Cheyenne and Board of Public Utilities Standards and Specifications.
- WYDOT Standard Specifications for Road and Bridge Construction (structure only).
- City of Cheyenne Unified Development Code (UDC).

Conflicts between design and construction standards will be resolved by the City.

A.6.b Surveying

Right-of-way surveying will be completed for the extent of the project as needed to complete the project design. Project control points will be established as needed. All surveying, design, and temporary control points will be referenced in the City control network. Property corners necessary to establish the right-of-way will be monumented as necessary. Existing right-of way, any required permanent and temporary easements, and parcel ownership will be shown in the project plan sheets.

A.6.c Geotechnical Investigation

A geotechnical investigation will be performed within the project limits and will include as necessary, but not be limited to:

- Geological strata defining soil and/or bedrock locations and elevations.
- Soil and/or rock properties.
- Soil classification.
- Recommended retaining wall and foundation types for any structures on the project (LRFD values).
- Subsurface drainage characteristics.
- Groundwater location.
- Any other characteristics not explicitly stated herein but required for completion of the project.

The results of the geological investigation and recommendations will be summarized in a geotechnical report. All required data, testing, bore hole locations, results, and recommendations, for the aforementioned investigations, will be summarized and included in the report. See detailed "Geotechnical Findings Evaluation" scope on the following pages for additional details.

A.6.d Surfacing Recommendations

Surfacing recommendations will be prepared based on geotechnical investigations performed throughout the project limits. The recommendations will outline surfacing thicknesses and recommended materials for subbase, base, and finished grade courses. Surfacing recommendations, material types, mix designs, and associated rates of application will be summarized in a pavement design report. All material types and rates of application will be included in the design plan summaries.

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A.6.e Utilities

All known utilities will be investigated and incorporated into the project design. These include, at a minimum:

- Telephone.
- Internet and cable.
- Electric and gas.
- Water and sanitary sewer.
- Signals, signs, and storm sewer.
- Fiber optic cable.

Any conflicts with other utilities, either active or abandoned, will be investigated and resolved.

Preliminary construction sequencing/phasing and traffic routing plans along with special provisions will be developed to facilitate utility work. Locates will be performed for any potential utility conflicts, this includes both surface utility locates and non-destructive subsurface utility excavation locates. All utilities within the project limits will be shown on the project plan sheets.

A.6.f Drainage Impact Study

A Drainage Impact Study will be performed for the project area in accordance with Section 3.2 of the Unified Development Code. The report will document the drainage impacts (if any), both upstream and downstream, of the project and present a design to mitigate any negative impacts. All supporting calculations will be included in the report. See detailed " Drainage Impact Study " scope on following pages for additional details.

A.6.g Project Reconnaissance Report

A preliminary reconnaissance report will be included as a supplement to this scope of work. After a consultant has been selected the report will be finalized by the consultant. This will require coordination with city staff and include a site walk-through. Any modifications or additions to the scope of work suggested by the consultant will be incorporated into the final report. The final reconnaissance report will be submitted and approved by city staff prior to design and plan submittals.

The report will also vet the preferred intersection and roadway for the location. A final recommendation of the most appropriate intersection and roadway configuration will be presented in the report, along with dialog outlining the final recommendation.

A.6.h Structure Selection Report

A structure selection memo will be included and will incorporate the most appropriate intersection configuration as determined in the reconnaissance report. The report will summarize potential structure alternatives; the hydraulic implications of each structure type with respect to roadway overtopping, freeboard requirements, and floodplain map changes; construction costs; lifecycle maintenance costs; and impacts to the travelling public. A final recommendation of the selection of the most appropriate structure

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will be presented in the report, along with dialog explaining the chosen alternative. See detailed "Structure Selection & Preliminary Design" scope on the following pages for additional details.

A.7 Plan Submittals

A.7.a General

Requirements for each formal plan issuance are outlined below and will act as a general guideline for submittals. All plan issuances will show utilities on plan and profile sheets. Additional intermediate and informal plan issuances may be required due any unforeseen issues encountered during the design process. It should be expected that there will be two rounds of reviews and one plan inspection meeting for each plan issuance.

A.7.b Preliminary Plans

Preliminary plans will be submitted illustrating the conceptual roadway alignment, intersection design, and structure layout. These plans should be approximately 30% complete. The general layout of the roadway, intersection, and structure should be clear and assist in the visualization and verification of the recommended intersection configuration, roadway alignment, and structure. At a minimum, this plan submittal will include the following design elements:

- Preliminary structure layout.
- Roadway and Greenway alignments.
- Intersection layout.
- Preliminary earthwork.
- Alternative design options that remain undecided.
- Preliminary Engineer's opinion of probable cost.

A.7.c 60% Plans

60% plans will be submitted containing the general layout of the roadway, intersection, and structures. These elements should be nearly finalized with only minor adjustments needed for utilities and right-of-way. At a minimum, the plans will include the following design elements:

- Roadway typical sections
- Vertical and horizontal roadway and Greenway alignments with plan and profile sheets.
- Identification of additional right-of-way needs (if applicable).
- Construction easements.
- Structure plan and elevations.
- Intersection layout and design.
- Earthwork.
- Estimated quantities.
- Engineer's opinion of probable costs.

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A.7.d 90% Plans

90% plans will be submitted when all right-of-way and utilities have been delineated and proposed relocations (if required) are finalized. All design elements required for the previous submittal will be finalized and incorporated into this plan set. This submittal will contain a complete set of construction documents, including plans, specifications, and estimates. This set of plans will contain the following:

- Title sheet.
- Legend sheet.
- Summary sheets, using City (roadway) and WYDOT (structure) bid items, which will include:
 - Individual summaries for all pay items.
 - Total Estimated Quantities summary.
- Roadway, Greenway, and structure typical sections.
- Roadway and Greenway Plan and Profile sheets.
- Intersection layout and details.
- Structure plan, elevations, and detail sheets.
- In addition to design plans the submittal will contain:
 - Design cross sections.
 - Earthwork.
 - Index of City of Cheyenne and Board of Public Utility Standard drawings.
 - Index of Standard WYDOT plans.
 - Index of Supplementary Specifications to the Standard Specifications.
 - Special Provisions to supplement the Standard Specifications.
 - Engineer's opinion of probable costs.

A.7.e Final Plans

All elements presented in previous plans will be finalized and all required changes from each respective plan issuance will be incorporated into this plan set. This submittal will contain a 100% complete set of construction documents, including plans, specifications, and estimates. All electronic design and plan files will be submitted to the City for archiving.

A.7.f Scope Additions and Clarifications

Intersection Evaluation and Recommendation

19th Street and Missile Drive:

The existing intersection was evaluated by the Cheyenne MPO in the summer of 2024, and it was determined that under projected 2045 traffic volumes, neither a traffic signal nor a 4-way stop control was warranted. Therefore, this will be a tee-intersection.

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Drainage Impact Study

Hydraulic analyses will be performed to determine what type of channel and structure clearance is needed to eliminate overtopping at 19th Street. Downstream impacts at Lincolnway and Ames Avenue Underpass will be evaluated in the analysis as well.

The Study and Report will include the following:

- Develop an existing conditions hydraulic model based on a direct site survey of the area of study and other available city (lidar) information. This will be a 2-dimensional HEC-RAS model.
- Three (3) general post-project conditions are anticipated to be modelled:
 - Condition 1; Base Flood Elevations (BFEs) in Crow Creek without a conveyance structure at 19th Street
 - Condition 2; BFEs with the existing conveyance at 19th Street
 - Condition 3; Includes up to three (3) bridge replacement options with the most detail on the recommended replacement conveyance and channel modification plan meeting the goals of the City for this project.
- Document and present findings to the city for review, including areas that deviate substantially from the effective floodplain delineations.
- Verify or reproduce tie-in points or other downstream features with known rating curves and the potential backwater influences.
- Present the required 'no-rise' documentation as applicable to the final design recommendations by the Design Team and accepted by the City. **Note that a CLOMR effort is included in this scope in the event a 'no-rise' solution is not feasible.**
- Remodelling and documentation of as-built conditions as applicable to the final project.

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STRUCTURAL AND ENVIRONMENTAL SCOPE OF WORK (HDR)

TASK SERIES 1.0 – General Conditions

HDR Tasks 1.1 through 1.6

- Task Series 1.0 General Conditions has been developed to capture overall project administration, project management, various project support tasks, and quality control costs throughout the duration of the project.
 - Project duration has been estimated at 9 months for the planning and design effort.
 - Assume attending one (1) on design review meeting per month (est. 2 person & 1 hr each)
 - Assume one (1) budget and schedule update per month (est. 2 people & 1 hr each)
 - Assume supplemental progress meetings via phone unless combined with other project trips per month as stated above
 - Assumes HDR participation in Project Reconnaissance effort with site visit attendance, providing comments and notes regarding the project structure and anticipated environmental impacts.
 - Assumes local HDR staff will support Ayres team with public involvement/meetings.
 - This consists of providing digital materials for graphics for use in meetings
 - Attendance of meetings with 2 local HDR staff

TASK SERIES 2.0 – Preliminary Design Phase

Task 2.1 – Environmental Resource Review

The following tasks indicate our team's approach to preparing the project for federal funding by completing a resource review that could be used for National Environmental Policy Act (NEPA) documentation in the future. Due to the involvement of FEMA Funding, HDR assumes that a FEMA NEPA document will be needed as part of the project. FEMA will be the lead that determines the type of document, so proposed is a two-step process: 1) Environmental Resource Review, and 2) NEPA document. The Environmental Resource Review is included in this scope and step 2, NEPA document would be completed as an amendment. The steps are to determine the type of document needed as part of the Environmental Resource Review. HDR will coordinate with FEMA to request their determination, as we have done on other projects for communities, to make sure FEMA is involved from the beginning, which saves back tracking to meet their requirements.

Desktop and field studies will be completed as part of the project in order to adequately evaluate resources within the Study Area. HDR will develop a study area that is 500-feet from the road centerlines for review and approval. This study area will be the basis for the resource review.

The resource review will be summarized in an environmental overview report which will document findings for later use in NEPA and assist on selection of build concepts. Resources of note in the study area include:

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Recreation Properties

Section 4(f) or Section 6(f) properties in the study area will be identified and documented. No further coordination is anticipated for this task.

Cultural Resources

HDR will prepare a summary of existing conditions for cultural resources within the study area. By conducting a file search of the Wyoming State Historic Preservation Office (SHPO) WyoTrack database to identify known cultural resources within the file search area. The file search area will include the direct study area and a 1-mile radius. Cultural resources within the file search area will be identified and summarized. National Register of Historic Places eligibility status for each cultural resource will be presented and used to identify known or potential historic properties within the study area.

Aquatic Resources

Aquatic Resources will be field delineated using the latest US Army Corps of Engineers (USACE) methodology. Data from the aquatic resource report will be used for the resource review and the 404 permitting in Task 3.2.

Biological Resources

A review of habitat for endangered and threatened species and other species of concern will be completed via windshield review of the study area. U.S. Fish and Wildlife's Information for Planning and Consultation (IPaC) website will be used to generate a list of threatened and endangered species and evaluate using IPaC determination keys. A USFWS coordination letter will be prepared based on the species list to request comments. No species-specific survey are included in this scope of work.

Socioeconomic Resources and Land Use

A review of socioeconomic resources that could be affected and potential land use changes resulting from construction of the project will be identified for inclusion in the Categorical Exclusion and Environmental Commitment Checklist.

Floodplains

The project crossed the Crow Creek. Information from the hydraulics analysis and structure design will be included in the resource review to determine possible impacts and any future permit needs.

Task 2.2 – Resource Overview Report

The following elements will be included in an environmental overview report which will summarize the environmental resources for future use in NEPA. The report will include:

- A concise Purpose and Need
- Alternatives considered.
- Summary of resources reviewed in Task 2.1.

Agency coordination will be prepared for agency review and comment to advise on the environmental concerns in the area, minimization measures, and possible future permitting.

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Deliverables:

- WY SHPO WYOTrack Record Search
- IPaC generated species list
- Aquatic Resources Delineation Report
- Agency Coordination Letter and responses
- Draft and Final Environmental Overview Report

Environmental Resource Overview Assumptions

- *HDR assumes the summary of existing conditions will not include reviews of General Land Office (GLO) maps, historical maps and aerial imagery, Sanborn Fire Insurance maps, Laramie County assessor records, or other archival sources.*
- *HDR assumes that the study area will not exceed 15 acres in size.*
- *HDR will prepare draft agency letters as word documents and will send out the letters to the applicable agencies as determined necessary.*
- *HDR assumes that the study will not require assessing potential effects to the Cheyenne West Historic District (48LA1418).*
- *Ayres will provide any pertinent project or design information to HDR for completion of the ROW and Utility portions of the CE checklist as needed.*
- *Assumes no standing structure relocations will be required for the project. Standing structure relocations may warrant an environmental assessment (a higher level of NEPA documentation), which would be handled with an amendment or separate contract.*
- *Changes to the general project alignment may warrant an amendment to this scope of services.*
- *This effort will not result in a NEPA decision. Once Federal funds are acquired and a lead federal agency has been identified, the class of NEPA documentation will be determined and completed under the direction of the federal agency.*

Task 2.3 – Structure Selection & Preliminary Design

- HDR will conduct preliminary bridge design and develop design alternatives to determine cost-effective bridge solutions as discussed in the RFP.
- Two bridge alternatives (Steel and Prestressed Girder) may be developed to help evaluate structure type, size, and conceptual level cost estimate.
 - A structure comparison memo will be provided to document the selected structure type.
- The City of Cheyenne has identified that the preferred structure will be a bridge type structure.
 - Specifically, it will be either a 2 or 3 span steel or prestressed girder bridge with a cast in place concrete deck.
 - The structure comparison memo will document and compare one potential alternate structure, such as buried structure, should it be applicable. However it is assumed that a structure of this type will not be chosen based on input from the City.
- Structure Deliverables include:
 - Roadway Alignment and Profile
 - Structure comparison memo and bridge Preliminary Layout plans
 - No design of retaining walls is anticipated during Preliminary Design phase

Task 2.4 – Geotechnical Findings Evaluation

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- Drilling and Lab Testing *(the following has been provided for the Prime to provide to the Geotechnical Engineering Subconsultant)*
 - The anticipated boring plan consists of two (2) geotechnical borings to be sampled to an approximate total depth of 50 feet below existing grade. Final depth will be dictated by the conditions encountered; however, a minimum of ten (10) feet of rock coring will be required for termination. The drilling and laboratory testing subcontractor will be expected to perform soil sampling utilizing the Standard Penetration Test (SPT) utilizing a 140-pound hammer. Soil sampling should be performed at 2.5-foot intervals to ten (10) feet below existing grade, followed by 5-foot intervals till auger refusal or total depth. Soil samples shall be collected in glass jars and stored for future laboratory testing. Rock cores shall be boxed in core boxes for future testing. It is assumed no contaminated soils or groundwater will be present, and Level D PPE will be required.
 - Upon completion of the borings, groundwater levels shall be collected if encountered. Borings shall be backfilled to approximately 0.5 feet from the surface, with a patch in kind at the surface. Drill cuttings may be used as backfill. The contractor is responsible for maintaining the backfilled holes such that there is no settlement of the patch.
 - The drilling contractor is responsible for Maintenance of Traffic (MOT) at the site in accordance with WYDOT standards. No time restrictions are anticipated.
- Geotechnical Findings Evaluation
 - HDR understands that the Prime will subcontract a drilling firm to perform the geotechnical borings for the proposed structure. HDR personnel will develop a recommended boring location plan with recommended testing, as needed based on site conditions. Prior to mobilizing for the fieldwork, a meeting will be held to discuss the proposed sampling and testing requirements. HDR personnel will coordinate with the subcontractor to be on site during the drilling operations, to log the samples, and evaluate the conditions of the site. Upon completion of the borings, HDR will review the boring logs and provide a recommended laboratory testing program to the subcontractor. Laboratory testing may consist of USCS classifications, shear strength testing, unidirectional consolidation, and unconfined compressive strength of rock. The actual laboratory testing program will be dictated by the conditions encountered in the borings. HDR will prepare geotechnical boring logs to be included in the foundation report.
 - HDR assumes the boring program will consist of two (2) borings at the project site, roughly in the opposite corners of the proposed structure, and will consist of Standard Penetration Testing (SPT) through the soil strata, followed by NQ-sized coring to depth. Assumed total depth of the borings is 50 feet, including ten (10) feet of rock coring. Maintenance of Traffic (MOT) will be handled by the subcontractor. We assume no time restrictions for drilling operations, and therefore assume a total of three (3) days on site.

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Project Name: 19th Street Drainage and Intersection Improvements Project

TASK SERIES 3.0 – Final Design Phase

Task 3.1 – Structural Design (Bridge)

- The selected bridge alternative will be designed according to the 9th Ed. of the AASHTO LRFD Bridge Design Specifications, 2020 and the Wyoming Bridge Design Manual. The design live load will be HL-93
- Below is a list of anticipated bridge design items
 - Determine and refine the bridge geometry including length, width, and elevations.
 - Develop the bridge loading requirements, including the level of seismic analyses required
 - Design bridge superstructure including bridge girders and reinforced concrete deck
 - Design bridge crash barriers and any required bridge railing(s)
 - Evaluate need and design for bridge expansion joints and bearings
 - Design bridge substructure including abutments and piers and their foundations
 - Evaluate the bridge foundations for the scoured condition
- Quality Assurance/Quality Control Review
 - HDR will review documents, specifications, assumptions, and calculations for accuracy and reasonableness.
 - QA/QC comments, responses, and resolutions for structure calculations, and special provisions will be documented and kept on file.
 - QA/QC Review will be completed by an experienced employee with HDR that has had limited involvement with the project and can review the project from a fresh perspective
- Assumptions
 - The fee associated with this proposal is based on a bridge type structure being chosen for the replacement.
- HDR will submit to the Wyoming Bridge Program per Wyoming statute 24-1-132 the following information:
 - To meet Wyoming Statute and federal requirements the following information is required to be submitted within ninety (90) days of the bridge being opened to traffic:
 - Design calculations, plans, and specifications
 - Load rating for all WYDOT Load Rating Trucks to include the AASHTO and WYDOT standard vehicles and the AASHTO EV2 and EV3 trucks following the Wyoming Bridge Design Manual and the AASHTO Manual for Bridge Evaluation 3rd Ed
 - Geotechnical report
 - Hydraulic report. The report will need to include scour elevations for the design, 100 year, and 500-year events
 - All submittals will be stamped by a Wyoming registered professional engineer

Task 3.2 – Geotechnical Recommendations (Structural)

- Geotechnical Design Memorandum and Engineering Geology Sheets
 - HDR personnel will prepare a Final Geology Report, for structural related items, summarizing the geotechnical conditions at the site and will include foundation recommendations, construction considerations, and design limitations. HDR will provide design recommendations per AASHTO design standards in the current LRFD Bridge Design Specifications and per WYDOT standards. The report will include borings logs, laboratory testing results, design calculations and references, special provisions, if needed, and the Engineering Geology sheets. The memo will be submitted electronically, and HDR assumes one (1) round of comments from the Owner.

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Task 3.3 – 404 Permit Application

- HDR will prepare a 404 Permit Application to the USACE evaluating overall project impacts to wetlands and/or waters of the US, potential cultural resources, and evaluate for threatened and endangered species. The following assumptions have been made in relation to this task:
 - All work required can be permitted under a Nationwide Permit.
 - No wetland mitigation will be required for the project. If wetland mitigation is deemed to be required, HDR can complete under additional scope.
 - No historic properties or archaeological sites are present within the study area.
 - No effect to any listed threatened or endangered species potentially present within the study area.
 - Habitat survey to confirm presence or absence of potential threatened and endangered species habitat will be conducted for the study area. Species specific or protocol surveys including fish surveys, or a Biological Assessment are not included in this scope, but if required may be negotiated as additional scope.
 - No additional permits for local or state agencies are included in this scope.
- The following item(s) will be provided as deliverable(s):
 - 404 Permit Application – Electronic Copy
- The following assumptions have been made:
 - Aquatic Resources Inventory would be completed by two (2) staff during the growing season (typically May – September), dependent on weather.

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“EXHIBIT B”

COMPENSATION SCHEDULE

The Consultant shall prepare and submit proposed estimated compensation, bearing in mind that it should be a unit-based contract with a negotiated not-to-exceed amount. A Payment Schedule based upon the following and any other pertinent payment schedule shall be submitted with payment for a specific phase not exceeding the scheduled amount prior to completion of that phase:

Data Gathering	\$ 191,929 .
Preliminary Plans	\$ 155,915 .
60% Plans	\$ 213,068 .
90% Plans	\$ 109,034 .
Final Plans	\$ 42,500 .
Total Contract Amount	\$ 712,446 .

Include this page as a part of the 19TH STREET DRAINAGE AND INTERSECTION IMPROVEMENTS PRICE PROPOSAL submission.

NOTE:

The above Cost Proposal includes \$25,120 in CLOMR effort in the event a No-Rise Condition cannot be achieved. If a No-Rise condition can be achieved, the CLOMR effort will not be invoiced. This fee is represented in the Final Plans Phase cost.



19th Street Drainage Intersection Improvements Fee Estimate

Project Manager: Nathan Silberhorn
Client: City of Cheyenne

Task #	Task Name	Total Fee	Total Direct Costs	Costs Markup	Sub-Consultant	Direct Costs	Total Labor Costs	APC	Labor Costs	Total Hours	Senior Prof. IV	Senior PM	Prof. III	Prof. II	Prof. I	Senior Designer	Admin. Assistant	SUE Supervisor	SUE Lead	SUE Technician
Project Plan		Fee								Hours										
Rates				0%				0.00%			\$200.00	\$195.00	\$165.00	\$145.00	\$125.00	\$135.00	\$95.00	\$150.00	\$135.00	\$110.00
		\$712,446	\$416,316	\$0	\$407,912	\$8,404	\$296,130	\$0	\$296,130	1823	16	268	280	433	524	40	31	51	92	104
Task 0	Contracting	\$4,361	\$126	\$0	\$0	\$126	\$4,235	\$0	\$4,235	26	0	12	0	8	0	0	3	3	0	0
0.1	Scope & Fee Negotiation	\$2,113	\$63	\$0	\$0	\$63	\$2,050	\$0	\$2,050	12		6		4				2		
0.2	Contracting	\$2,248	\$63	\$0	\$0	\$63	\$2,185	\$0	\$2,185	14		6		4			3	1		
Task 1	Kickoff Meeting & Project Management	\$28,523	\$693	\$0	\$0	\$693	\$27,830	\$0	\$27,830	170	0	82	0	52	0	0	20	16	0	0
1.1	Project Kick-off Meeting	\$2,213	\$63	\$0	\$0	\$63	\$2,150	\$0	\$2,150	14		4		4			2	4		
1.2	Site Reconnaissance & Report	\$2,933	\$63	\$0	\$0	\$63	\$2,870	\$0	\$2,870	17		8		8				1		
1.3	Progress Meetings	\$13,877	\$567	\$0	\$0	\$567	\$13,310	\$0	\$13,310	84		40		20			18	6		
1.4	Budgeting & Invoicing	\$4,150	\$0	\$0	\$0	\$0	\$4,150	\$0	\$4,150	25		10		10				5		
1.5	Management of Disciplines	\$5,350	\$0	\$0	\$0	\$0	\$5,350	\$0	\$5,350	30		20		10						
Task 2	Topographic Survey	\$20,550	\$20,550	\$0	\$20,550	\$0	\$0	\$0	\$0	0	0	0	0	0	0	0	0	0	0	0
2.1		\$20,550	\$20,550	\$0	\$20,550	\$0	\$0	\$0	\$0	0										
Task 3	Geotechnical Investigation	\$27,100	\$27,100	\$0	\$27,100	\$0	\$0	\$0	\$0	0	0	0	0	0	0	0	0	0	0	0
3.1	Drilling & Testing - 19th St Structural	\$16,000	\$16,000	\$0	\$16,000	\$0	\$0	\$0	\$0	0										
3.2	Drilling, Testing, Engineering - 19th St Roadway	\$9,600	\$9,600	\$0	\$9,600	\$0	\$0	\$0	\$0	0										
3.3	Pavement Recs	\$1,500	\$1,500	\$0	\$1,500	\$0	\$0	\$0	\$0	0										
Task 4	Environmental Investigation & Clearance	\$21,046	\$21,046	\$0	\$21,046	\$0	\$0	\$0	\$0	0	0	0	0	0	0	0	0	0	0	0
4.1	Environmental Clearances	\$11,586	\$11,586	\$0	\$11,586	\$0	\$0	\$0	\$0	0										
4.2	Resource Overview Report	\$4,540	\$4,540	\$0	\$4,540	\$0	\$0	\$0	\$0	0										
4.3	404 Permit Application	\$4,920	\$4,920	\$0	\$4,920	\$0	\$0	\$0	\$0	0										
Task 5	Public Involvement	\$10,974	\$4,154	\$0	\$4,154	\$0	\$6,820	\$0	\$6,820	40	0	26	0	0	14	0	0	0	0	0
5.1	Public Meeting 1	\$4,085	\$1,385	\$0	\$1,385	\$0	\$2,700	\$0	\$2,700	16		10			6					
5.2	Public Meeting 2 (30%)	\$3,445	\$1,385	\$0	\$1,385	\$0	\$2,060	\$0	\$2,060	12		8			4					
5.3	Public Meeting 3 (60%)	\$3,445	\$1,385	\$0	\$1,385	\$0	\$2,060	\$0	\$2,060	12		8			4					
Task 6	Subsurface Utility Engineering	\$44,870	\$6,610	\$0	\$0	\$6,610	\$38,260	\$0	\$38,260	276	16	0	0	0	8	40	0	32	92	104
6.1	Field Locating & Designating	\$11,400	\$0	\$0	\$0	\$0	\$11,400	\$0	\$11,400	88	2							8	40	40
6.2	Utility Report & Matrix	\$5,550	\$0	\$0	\$0	\$0	\$5,550	\$0	\$5,550	38	4							8	10	20
6.3	Utility Plans	\$8,710	\$0	\$0	\$0	\$0	\$8,710	\$0	\$8,710	62	2				8	40		8	2	4
6.4	SUE QL-A (Assumes 10 Test Holes)	\$19,210	\$6,610	\$0	\$0	\$6,610	\$12,600	\$0	\$12,600	88	8							8	40	40
Task 7	Traffic Operations Evaluation	\$15,325	\$0	\$0	\$0	\$0	\$15,325	\$0	\$15,325	105	0	14	0	73	10	0	8	0	0	0
7.1	Warrant and Traffic Projection review	\$580	\$0	\$0	\$0	\$0	\$580	\$0	\$580	4				4						
7.2	Traffic Operations Evaluation	\$7,440	\$0	\$0	\$0	\$0	\$7,440	\$0	\$7,440	52		2		40	10					
7.3	Draft Traffic Analysis Memo	\$3,670	\$0	\$0	\$0	\$0	\$3,670	\$0	\$3,670	26		2		20			4			
7.4	Final Traffic Analysis Memo	\$1,495	\$0	\$0	\$0	\$0	\$1,495	\$0	\$1,495	11		2		5			4			
7.5	Intersection Selection	\$2,140	\$0	\$0	\$0	\$0	\$2,140	\$0	\$2,140	12		8		4						
Task 8	Floodplain Analysis	\$58,900	\$58,900	\$0	\$58,900	\$0	\$0	\$0	\$0	0	0	0	0	0	0	0	0	0	0	0
8.1	Data Gathering	\$9,640	\$9,640	\$0	\$9,640	\$0	\$0	\$0	\$0	0										
8.2	Preliminary Plans	\$13,580	\$13,580	\$0	\$13,580	\$0	\$0	\$0	\$0	0										
8.3	60% Plans	\$3,980	\$3,980	\$0	\$3,980	\$0	\$0	\$0	\$0	0										
8.4	90% Plans	\$2,080	\$2,080	\$0	\$2,080	\$0	\$0	\$0	\$0	0										
8.5	Final Plans	\$4,500	\$4,500	\$0	\$4,500	\$0	\$0	\$0	\$0	0										
8.6	CLOMR	\$25,120	\$25,120	\$0	\$25,120	\$0	\$0	\$0	\$0	0										
Task 9	Bridge Selection & Design	\$276,163	\$276,163	\$0	\$276,163	\$0	\$0	\$0	\$0	0	0	0	0	0	0	0	0	0	0	0
9.1	General Conditions	\$31,143	\$31,143	\$0	\$31,143	\$0	\$0	\$0	\$0	0										
9.2	Preliminary Design	\$44,440	\$44,440	\$0	\$44,440	\$0	\$0	\$0	\$0	0										
9.3	Final Design	\$200,580	\$200,580	\$0	\$200,580	\$0	\$0	\$0	\$0	0										
Task 10	Urban Drainage Design	\$26,574	\$244	\$0	\$0	\$244	\$26,330	\$0	\$26,330	212	0	0	0	68	44	56	0	0	0	0
10.1	Basin & System Review	\$2,714	\$244	\$0	\$0	\$244	\$2,470	\$0	\$2,470	20				8	8					
10.2	Hydraulics Modeling	\$6,530	\$0	\$0	\$0	\$0	\$6,530	\$0	\$6,530	52				30	10					
10.3	Pipe Network	\$4,710	\$0	\$0	\$0	\$0	\$4,710	\$0	\$4,710	40				10	10	8				
10.4	30% Plans	\$5,480	\$0	\$0	\$0	\$0	\$5,480	\$0	\$5,480	44				8	8	20				
10.5	60% Plans	\$4,630	\$0	\$0	\$0	\$0	\$4,630	\$0	\$4,630	36				8	8	16				
10.6	90% Plans	\$1,970	\$0	\$0	\$0	\$0	\$1,970	\$0	\$1,970	16				4		8				
10.7	100% Plans	\$540	\$0	\$0	\$0	\$0	\$540	\$0	\$540	4						4				
Task 11	Roadway Design	\$49,720	\$0	\$0	\$0	\$0	\$49,720	\$0	\$49,720	356	0	26	0	170	160	0	0	0	0	0
11.1	Horizontal and Vertical Design	\$4,340	\$0	\$0	\$0	\$0	\$4,340	\$0	\$4,340	32		2		10	20					
11.2	Intersection Layouts & Details	\$11,970	\$0	\$0	\$0	\$0	\$11,970	\$0	\$11,970	86		6		40	40					
11.3	Driveways & Access	\$9,070	\$0	\$0	\$0	\$0	\$9,070	\$0	\$9,070	66		6		20	40					
11.4	Sidewalks & Ramps	\$12,360	\$0	\$0	\$0	\$0	\$12,360	\$0	\$12,360	88		8		40	40					
11.5	Control Points	\$5,790	\$0	\$0	\$0	\$0	\$5,790	\$0	\$5,790	42		2		20	20					
11.6	3D Modeling	\$6,190	\$0	\$0	\$0	\$0	\$6,190	\$0	\$6,190	42		2		40						
Task 12	Plans, Specifications, & Estimate Development	\$128,341	\$731	\$0	\$0	\$731	\$127,610	\$0	\$127,610	850	0	108	280	130	332	0	0	0	0	0
12.1	30% Design Plans & Costs	\$37,120	\$0	\$0	\$0	\$0	\$37,120	\$0	\$37,120	256		16	80	40	120					
12.2	30% Design Review Meeting	\$2,184	\$244	\$0	\$0	\$244	\$1,940	\$0	\$1,940	12		4	4		4					
12.3	60% Design Plans, Costs, & Specs	\$40,770	\$0	\$0	\$0	\$0	\$40,770	\$0	\$40,770	266		36	100	50	80					
12.4	60% Design Review Meeting	\$2,184	\$244	\$0	\$0	\$244	\$1,940	\$0	\$1,940	12		4	4		4					
12.5	90% Design Plans, Costs, & Specs	\$36,020	\$0	\$0	\$0	\$0	\$36,020	\$0	\$36,020	236		36	80	40	80					
12.6	90% Design Review Meeting	\$2,184	\$244	\$0	\$0	\$244	\$1,940	\$0	\$1,940	12		4	4		4					
12.7	Final Design Plans, Costs, & Specs	\$7,880	\$0	\$0	\$0	\$0	\$7,880	\$0	\$7,880	56		8	8		40					
Task 13	Construction Administration & Observation	TBD								0	0	0	0	0	0	0	0	0	0	0



Project: 19th Street Drainage and Intersection
Improvements Project
Subject: Compensation Summary
Task: Contracting

Computed by: HDR Cheyenne Date: 8/2/2024
Workbook: Comp.Sched.
Page: 1 of 1

PROFESSIONAL SERVICES - DESIGN			
Task	Labor	Expenses	Total
Task 1.0 General Conditions	\$ 34,875.00	\$ 420.50	\$ 35,295.50
Task 2.0 Preliminary Design	\$ 59,180.00	\$ 1,386.00	\$ 60,566.00
Task 3.0 Final Design	\$ 205,330.00	\$ 170.00	\$ 205,500.00
Subtotals		\$ 299,385.00	\$ 1,976.50
Design Total Not-to-Exceed			\$ 301,361.50
TOTAL PROJECT		\$	301,361.50

Level of Effort and Fee Estimate
Ayres Associates / City of Cheyenne, Wyoming
19th Street Drainage and Intersection Improvements

	Project Team																Expenses								
Work Task	Prj. Principal	QA/QC	PM	Engineer III	Structural Lead	Structural Eng. III	Structural Eng. I	Structural CAD	Sr. Geologist	Geotech. Eng. W.	Environmental Sci.	Environmental Sci.	Cultural Resources	Project Admin	Clerical	HDR Labor Total	Meal						Total Direct Expenses		Total Project
	M.Oakley	D.Hamilton	T.Mattson	B.Collamer	T. Borchard	L. Rogers	B.Winter	S. Miller	D. Crotsley	Brandenberger	J. Rust	M. Carlson	P. Buckner	E. Schramm	C. Henze		Mileage	Lodging	Per Diem	Lab Testing	Copies	WYOTrack Records Search	GPS		
																	Mile	Night	\$/Day			\$/Tot	Hour		
Contract Billing Rate	\$ 205.00	\$ 205.00	\$ 190.00	\$ 145.00	\$ 225.00	\$ 145.00	\$ 120.00	\$ 120.00	\$ 200.00	\$ 160.00	\$ 180.00	\$ 110.00	\$ 125.00	\$ 95.00	\$ 95.00		\$ 0.67	\$ 150.00	\$ 59.00	\$ 1.00	\$ 1.00	\$ 780.00	\$ 50.00		
1.0 General Conditions	8	8	68	36	22	23	9	2	0	0	1	4	0	16	18	\$34,875	150	0	0	0	320	0	0	\$421	\$35,296
1.1 Project Management	6		24		10	23	9	2							4	\$13,075.00	50				20			\$ 53.50	
1.2 Quality Control		8	4												4	\$2,780.00								\$ -	
1.3 Budget & Schedule Updates	2		10	10										16	6	\$5,850.00								\$ -	
1.4 Progress Meetings			10	10	12											\$6,050.00								\$ -	
1.5 Project Reconnaissance Report			8	8							1	4				\$3,300.00	50							\$ 33.50	
1.6 Public Involvement Support			12	8											4	\$3,820.00	50				300			\$ 333.50	
																\$0.00									
2.0 Preliminary Design	2	4	11	4	32	52	90	20	20	56	8	104	12	0	0	\$59,180	1000	5	7	1500	70	1	0	\$1,386	\$60,566
2.1 Environmental Clearances											4	80	6			\$10,270.00		2	4		1			\$ 1,316.00	
2.2 Resource Overview Report			2								4	24	6			\$4,490.00					50			\$ 50.00	
2.3 Structure Selection & Preliminary Design	2		8	4	32	52	90	20								\$30,450.00					20			\$ 20.00	
2.4 Geotechnical Findings Evaluation		4	1						20	56						\$13,970.00	1,000	3	3	1,500.00				\$ 2,797.00	
																\$0.00									
3.0 Final Design	2	10	8	8	62	368	571	349	28	80	8	24	0	0	0	\$205,330	0	0	0	0	170	0	0	\$170	\$205,500
3.1 Structural Final Design (Bridge & Ret.Wall)	2	8	4	8	62	368	571	345								\$181,200.00					100			\$ 100.00	
3.2 Geotechnical Recommendations (Structural)			2					4	28	80						\$19,260.00					20			\$ 20.00	
3.3 404 Permit Application		2	2								8	24				\$4,870.00					50			\$ 50.00	
																\$0.00								\$ -	
Total Labor Hours per Team Member	12	22	87	48	116	443	670	371	48	136	17	132	12	16	18	299,385	1,150	5	7	1,500	560	1	0	\$ 1,977	
Total Labor Charges per Team Member	\$ 2,460	\$ 4,510	\$ 16,530	\$ 6,960	\$ 26,100	\$ 64,235	\$ 80,400	\$ 44,520	\$ 9,600	\$ 21,760	\$ 3,060	\$ 14,520	\$ 1,500	\$ 1,520	\$ 1,710										
Total Project																									\$ 301,362

CivilWorx Fee Schedule

19th Street Drainage and Intersection - 2024

		Principal	Project Manager / Senior Engineer	Professional Engineer	Associate Engineer	Clerical/CAD/GIS	CivilWorx, LLC	Office/Printing/Mailings	Reimbursable Expenses	Totals
Hourly Rate:		\$150	\$135	\$105	\$85	\$80				
Task No. and Description		Hrs.	Hrs.	Hrs.	Hrs.	Hrs.				
Task 1	Data Gathering									
1.1	Design, Meetings, Coordination		8	8			\$1,920	\$ -		\$1,920
1.2	Model Development, Initial Report		24	24	8	16	\$7,720	\$ -		\$7,720
Subtotal Task 1		-	32	32	8	16	\$9,640			\$9,640
Task 2	Preliminary Plans									
2.1	Meetings, Reports, and Corrdination		16	8	4	8	\$3,980			\$3,980
2.2	Modeling and Design Iterations		40	40			\$9,600			\$9,600
Subtotal Task 2		-	56	48	4	8	\$13,580			\$13,580
Task 3	60% Plans									
3.1	Meetings, Reports, and Corrdination		16	8	4	8	\$3,980			\$3,980
Subtotal Task 3		-	16	8	4	8	\$3,980			\$3,980
Task 4	90% Plans									
4.1	Meetings, Reports, and Corrdination		8	8		2	\$2,080			\$2,080
Subtotal Task 4		-	8	8	-	2	\$2,080			\$2,080
Task 5	Final Plans									
5.1	Meetings, Reports, and Corrdination		16	16	4	4	\$4,500			\$4,500
Subtotal Task 5		-	16	16	4	4	\$4,500			\$4,500
PROPOSAL TOTALS		-	128.0	112.0	20.0	38.0	33,780.0	-	-	33,780.0

Optional Additional Fee

Task 6	CLOMR									
6.1	Application Fee & Effective Model Request Fee						\$ -	\$7,000		\$7,000
6.2	Application Paperwork and Report		16	24			\$4,680	\$ -		\$4,680
6.30	Model Development and Submittal		16	16		4	\$4,160			\$4,160
6.4	Workmaps and Annotative FIRM		8	8		40	\$5,120			\$5,120
6.5	Review Comments through acceptance		16	16		4	\$4,160			\$4,160
Subtotal Task 6 (optional)		-	56	64	-	48	\$18,120		\$7,000	\$25,120

From: COTTON JONES <cotton.jones@msn.com>
Sent: Monday, July 8, 2024 12:39 PM
To: Silberhorn, Nathan
Subject: RE: Cheyenne 19th Street intersection survey

Nathan,

Please see below estimate for the 19th Street project survey.

1. Drone Photogrammetry & LiDAR Survey - \$8,000
 - a. Set ±6 aerial targets for drone survey
 - b. Drone scan with a mapping camera
 - c. ECW Image File loaded into CAD
 - d. TIN Surface for areas outside road and adjacent parking lots
2. Surface Extraction with Conventional- \$2,380
 - Adding conventional survey data to drone surface
3. Planimetric Drafting- \$3,170
 - Planimetric Drafting - Surface linework at Client's standard layers
4. Conventional Survey Support – \$7,000
 - GPS & Robotic Total Station to support drone mapping of hard surfaces including curb & gutter, street surfaces, sidewalks, pipe inverts, etc.
 - Boundary determination for street right-of-ways.
 - Establish Project Control

Estimated cost for survey services is \$20,550

Look it over and let me know if I missed anything.

Thanks,

Cotton D. Jones, P.L.S., CFedS



Please consider the environment before printing this e-mail

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Terracon Fee Estimate - ROADWAY GEOTECHNICAL SCOPE
Cheyenne 19th Street Drainage & Intersection

Number of Borings	Planned Boring Depth (feet) ¹	Planned Location ²
3	10	Within the existing pavement limits of Missile Drive

Task	Lump Sum Fee ^{1,2}
Boring Layout and Private Utility Locating Services ³	\$1,830
Field Exploration ⁴	\$2,940
Laboratory Testing	\$1,690
Geotechnical Analysis and Reporting	\$2,230
Project Management	\$910
Sub Total	\$9,600

1. Proposed fees noted above are effective for 90 days from the date of the proposal.
2. Proposed fees noted above will be invoiced on a per month basis and/or after completion of each task.
3. Includes Terracon personnel time onsite while marking and clearing boring locations with private utility locator.
4. The lump sum fee considers one drill rig mobilization and no unexpected onsite delays. If additional drill rig mobilizations are required, an additional fee of \$1,250 would be invoiced. A drill crew standby rate of \$500 per hour would be invoiced for unexpected delays.

Task	Lump Sum Fee ^{1,2}
Pavement Recommendations and Report	\$1,500
Total	\$11,100

Exhibit B - STRUCTURE GEOTECHNICAL SCOPE

Our proposed Scope of Services consists of field exploration, laboratory testing, and project delivery. Our Scope of Services does not include providing geotechnical engineering recommendations or design criteria for the project. These services are described in the following sections.

Field Exploration

Based on input provided by Ayres, and our experience with similar projects in the vicinity of the project site, we propose the following field exploration program as described below:

Number of Borings	Planned Boring Depth (feet) ¹	Planned Location ²
2	50	Within the existing pavement limits of Missile Drive

1. Below existing ground surface. **If sufficient subsurface information is not obtained to provide design-level recommendations after termination of planned boring depth, additional exploration and mobilization of specialty equipment may be required. Cost for specialty equipment mobilization and drilling, such as rock coring, will be provided upon your request.**
2. We understand Ayres will provide boring locations prior to our mobilization to the site.

Pre-Exploration Site Visit: Prior to mobilizing our drilling equipment to the project site, a Terracon representative will visit the site to meet with a representative of the Ayres and/or the City of Cheyenne (as needed) to coordinate our field work, stake/mark boring locations, assess drill rig access/regress to/from the boring locations, and confirm public and private utilities have been appropriately marked or cleared. If conditions are identified during our pre-mobilization site visit that could negatively impact the field exploration, relative to private utilities and/or equipment access, we will contact you to discuss our concerns and options, and additional fees may apply.

Boring Layout and Elevations: We will coordinate with your Project Surveyor to include locations and surface elevations in project information.

Subsurface Exploration Procedures: We will advance the soil borings with a truck-mounted drill rig using continuous-flight augers (solid-stem and/or hollow-stem, as necessary, depending on subsurface conditions). Typically, four samples are obtained in the upper 10 feet of each boring and at intervals of 5 feet thereafter. Sampling will be in

Proposal for Geotechnical Engineering Services

19th Street Drainage and Intersection Improvements | Cheyenne, Wyoming
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general accordance with local standard procedures wherein modified California barrel or split-barrel samples are obtained. In the split barrel sampling procedure, a standard 2-inch outer diameter split barrel sampling spoon is driven into the ground by a 140-pound automatic hammer falling a distance of 30 inches. The number of blows required to advance the sampling spoon the last 12 inches of a normal 18-inch penetration is recorded as the Standard Penetration Test (SPT) resistance value. The SPT resistance values, also referred to as N-values, are indicated on the boring logs at the test depths. A 2.5-inch O.D. split-barrel modified California sampling spoon with 2.0-inch I.D. tube lined sampler may also be used for sampling. The modified California barrel sampling procedures are similar to standard split spoon sampling procedure; however, blow counts are typically recorded for 6-inch intervals for a total of 12 inches of penetration. Bulk samples of auger cuttings may also be obtained from select boring locations and depths. The samples will be placed in appropriate containers, taken to our soil laboratory for testing, and classified by a Geotechnical Engineer.

Groundwater levels will be observed and recorded during and after drilling operations. Provisions have not been made to obtain delayed groundwater readings.

Our exploration team will prepare field boring logs as part of standard drilling operations including sampling depths, penetration distances, and other relevant sampling information. Field logs include visual classifications of materials observed during drilling and our interpretation of subsurface conditions between samples. Final boring logs, prepared from field logs, represent the Geotechnical Engineer's interpretation, and include modifications based on observations and laboratory tests.

Property Disturbance: Terracon will make reasonable efforts to reduce damage to the property. However, it should be understood that in the normal course of our work some disturbance could occur including rutting of the ground surface and damage to landscaping and/or pavements.

We will backfill borings with auger cuttings mixed with bentonite chips and/or pea gravel upon completion. Pavements will be patched with asphalt and/or concrete, as appropriate. Our services do not include repair of the site beyond backfilling our boreholes and patching existing pavements/surfaces. Excess auger cuttings will be removed from the roadway. Because backfill material often settles below the surface after a period, we recommend boreholes to be periodically checked and backfilled, if necessary.

Safety

Terracon is not aware of environmental concerns at this project site that would create health or safety hazards associated with our exploration program; thus, our Scope considers standard OSHA Level D Personal Protection Equipment (PPE) appropriate. Our



Scope of Services does not include environmental site assessment services, but identification of unusual or unnatural materials observed while drilling will be noted on our logs and discussed in our report.

If suspected environmental impacts are observed or encountered during the field exploration, Ayres will be notified. The potential of environmental impact to field program represents a change in condition for the project. The environmental services related to continuing the field exploration are not included in this scope or services. Terracon is available to provide these services as a supplement to this proposal.

Exploration efforts require borings into the subsurface; therefore, Terracon will comply with local regulations to request a utility location service through Wyoming One-Call. We will consult with Ayres/City of Cheyenne regarding potential utilities or other unmarked underground hazards. Based upon the results of this consultation, we will consider the need for alternative subsurface exploration methods as the safety of our field crew is a priority.

Private utilities should be marked by the Ayres/City of Cheyenne and any information provided prior to commencement of field exploration. Terracon will not be responsible for damage to private utilities not disclosed to us, and we will not perform excavations without an understanding of the subsurface utilities present based upon markings made by the various responsible parties.

Due to the site being developed, **Terracon's safety policy requires private utility services to be performed at all developed sites prior to performing subsurface exploration activities. We will subcontract private utility locating services to identify any private utilities in the vicinity of the proposed borings to reduce the risk of a utility strike.** Fees for this service are included in our Scope of Services and can be found in **Exhibit C – Compensation and Project Schedule** of this proposal.

Please note that the detection of underground utilities is dependent upon the composition and construction of the utility line; some utilities are comprised of non-electrically conductive materials and may not be readily detected. The use of a private utility locate service would not relieve the landowner/client of their responsibilities in identifying private underground utilities.

Site Access and Right-of-Way Permit: Terracon must be granted access to the public street right-of-way by the City of Cheyenne. We will work diligently with you to obtain all required Right-of-Way (ROW) permit and other necessary permits. Our proposed fees assume all fees associated with permitting through the City of Cheyenne will be the responsibility of Ayres and/or City of Cheyenne. Terracon will conduct field services during normal business hours (Monday through Friday between 7:00am and 5:00pm). If



our exploration must take place over a weekend or at night, please contact us so we can adjust our schedule and fee.

Traffic Control: All our field operations, including performing private utility clearances at boring locations and drilling, will be conducted with the assistance of subcontracted professional traffic control services (S&J Signs). Fees for subcontracted traffic control services are provided in **Exhibit C – Compensation and Project Schedule** of this proposal.

Laboratory Testing

The project engineer will review field data and work with HDR to assign laboratory tests to understand the engineering properties of various soil strata. Exact types and number of tests cannot be defined until completion of fieldwork, but we anticipate the following laboratory testing may be performed:

- Water content
- Dry unit weight
- Atterberg limits
- Grain size analysis
- One-dimensional swell/collapse (if applicable)
- Moisture-density relationship
- California Bearing Ratio (CBR)
- Chemical analyses – pH, soluble sulfate, sulfides, soluble chloride, RedOx, electrical resistivity, and total salts

Our laboratory testing program includes examination of soil samples by an engineer/geologist. Based on the results of our field and laboratory programs, we will describe and classify soil samples in accordance with the Unified Soil Classification System (USCS).

Engineering and Project Delivery

The results of our field and laboratory programs will be evaluated, and a geotechnical data report will be prepared under the supervision of a licensed professional engineer. The geotechnical data report will provide the following:

- Site Location and Exploration Plans
- Subsurface exploration procedures
- Description of subsurface conditions
- Boring logs with:

Proposal for Geotechnical Engineering Services

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- Stratification based on visual soil (and rock) classification
- Summarized laboratory data
- Groundwater levels observed in the boreholes
- Approximate ground surface elevation
- Approximate latitude-longitude

In addition to an emailed data report, your project will also be delivered using the **Terracon Compass** website. Upon initiation, we provide you and your design team with the necessary link and password to access the website (if not previously registered). Each project includes a calendar to track the schedule, an interactive site map, a listing of team members, access to the project documents as they are uploaded to the site, and a collaboration portal. We welcome the opportunity to have project kickoff conversations with the team to discuss key elements of the project and demonstrate features of the portal. The typical delivery process includes the following:

- Project Planning – Proposal information, schedule, and anticipated exploration plan
- Site Characterization – Findings of the site exploration and laboratory results
- Geotechnical Data Report

When services are complete, we upload a printable version of our completed Geotechnical Data report, including the Professional Engineer's seal and signature, which documents our services. Previous submittals, collaboration, and the report are maintained in our system. This allows future reference and integration into subsequent aspects of our services as the project goes through final design and construction.

Additional Services

In addition to the services noted above, the following are often associated with geotechnical engineering services. Fees for services noted above do not include the following:

Environmental Assessments: Our Scope for this project does not include, either specifically or by implication, an environmental assessment of the site intended to identify or quantify potential site contaminants. If the client/owner is concerned about the potential for such conditions, an environmental site assessment should be conducted. We can provide a proposal for an environmental assessment, if desired.



Exhibit C - Compensation and Project Schedule

Compensation

Based upon our understanding of the site, the project as summarized in Exhibit A, and our planned Scope of Services described in Exhibit B, our base fee is shown in the following table:

Task	Lump Sum Fee ^{1,2}
Site Visit, Boring Layout and Private Utility Locating Services ³	\$2,845
Traffic Control and Field Exploration ⁴	\$7,370
Laboratory Testing	\$4,030
Geotechnical Data Reporting	\$1,090
Project Management	\$665
Total	\$16,000

1. Proposed fees noted above are effective for 90 days from the date of the proposal.
2. Proposed fees noted above will be invoiced on a per month basis and/or after completion of each task.
3. Includes Terracon personnel time onsite while marking and clearing boring locations with private utility locator.
4. The lump sum fee considers one drill rig mobilization and no unexpected onsite delays. If additional drill rig mobilizations are required, an additional fee of \$1,250 would be invoiced. A drill crew standby rate of \$500 per hour would be invoiced for unexpected delays.

Our Scope of Services does not include services associated with site clearing, wet ground conditions, tree or shrub clearing, or repair of/damage to existing landscape. If such services are desired by the owner/client, we should be notified so we can adjust our Scope of Services.

Unless instructed otherwise, we will submit our invoice(s) to the address shown at the beginning of this proposal. If conditions are encountered that require Scope of Services revisions and/or result in higher fees, we will contact you for approval, prior to initiating services. A supplemental proposal stating the modified Scope of Services as well as its effect on our fee will be prepared. We will not proceed without your authorization.

PROFESSIONAL SERVICES AGREEMENT

Project Name: 19th Street Drainage and Intersection Improvements Project

PROJECT SCHEDULE

The key milestones for the **19TH STREET DRAINAGE AND INTERSECTION IMPROVEMENTS Project.**

The proposer should complete the following schedule as part of the proposal and include any other key dates needed.

Schedule Item	Date
Notice-to-proceed issued by City	September 16, 2024
Project Kick-Off Meeting	<u>September 18, 2024</u>
Final Recon Report	<u>September 30, 2024</u>
Preliminary Plans	<u>December 17, 2024</u>
Structure Selection Report	<u>November 19, 2024</u>
60% Plans	<u>February 28, 2024</u>
90% Plans	<u>May 20, 2025</u>
Project Completion	June 30, 2025